US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12410079
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Cote; Pierre Lucien et al.

Membrane assembly for supporting a biofilm

Abstract

A cord for supporting a biofilm has a plurality of yarns. At least one of the yarns comprises a plurality of hollow fiber gas transfer membranes. At least one of the yarns extends along the length of the cord generally in the shape of a spiral. Optionally, one or more of the yarns may comprise one or more reinforcing filaments. In some examples, a reinforcing yarn is wrapped around a core. A module may be made by potting a plurality of the cords in at least one header. A reactor may be made and operated by placing the module in a tank fed with water to be treated and supplying a gas to the module. In use, a biofilm covers the cords to form a membrane biofilm assembly.

Inventors: **Cote; Pierre Lucien (Ancaster, CA), Pedersen; Steven Kristian (Nottawa, CA)**

Applicant: **BL TECHNOLOGIES, INC.** (Minnetonka, MN)

Family ID: **47884526**

Assignee: **BL TECHNOLOGIES, INC. (Minnetonka, MN)**

Appl. No.: **18/338182**

Filed: **June 20, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230331605 A1	Oct. 19, 2023

Related U.S. Application Data

continuation parent-doc US 16998704 20200820 US 11724947 child-doc US 18338182
continuation parent-doc US 14769461 US 10781119 20200922 WO PCT/US2013/027435 20130222 child-doc US 16998704

Publication Classification

Int. Cl.: **C02F3/12** (20230101); **B01D63/02** (20060101); **B01D63/04** (20060101); **B01D69/08** (20060101); **C02F3/10** (20230101); **C02F3/20** (20230101)

U.S. Cl.:

CPC **C02F3/1273** (20130101); **B01D63/027** (20130101); **B01D63/04** (20130101); **B01D69/081** (20130101); **C02F3/102** (20130101); **C02F3/103** (20130101); **C02F3/208** (20130101); B01D2325/40 (20130101); C02F2303/20 (20130101); Y02W10/10 (20150501)

Field of Classification Search

CPC: C02F (3/208); C02F (3/20); C02F (3/1273); B01D (2313/26)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2458163	12/1948	Hays	N/A	N/A
3226317	12/1964	Albertson	N/A	N/A
4066553	12/1977	Bardonnet	N/A	N/A
4067801	12/1977	Ishida	N/A	N/A
4126544	12/1977	Baensch	N/A	N/A
4181604	12/1979	Ryozo	N/A	N/A
4270702	12/1980	Nicholson	N/A	N/A
4328102	12/1981	Bellhouse	N/A	N/A
4341005	12/1981	Oscarsson	N/A	N/A
4416993	12/1982	McKeown	N/A	N/A
4428403	12/1983	Lee	N/A	N/A
4563282	12/1985	Wittmann	N/A	N/A
4664681	12/1986	Anazawa	N/A	N/A
4746435	12/1987	Onishi	N/A	N/A
4883594	12/1988	Sekoulov	210/603	C02F 3/208
4960546	12/1989	Tharp	N/A	N/A
5015421	12/1990	Messner	N/A	N/A
5034164	12/1990	Semmens	N/A	N/A
5034165	12/1990	Willinger	N/A	N/A
5043140	12/1990	Combs	N/A	N/A
5116506	12/1991	Williamson	N/A	N/A
5126050	12/1991	Irvine	N/A	N/A
5149649	12/1991	Miyamori	N/A	N/A
5213685	12/1992	Padovan	N/A	N/A
5238562	12/1992	Rogut	N/A	N/A
5282964	12/1993	Young	N/A	N/A
5374138	12/1993	Byles	N/A	N/A
5395468	12/1994	Juliar	N/A	N/A
5439736	12/1994	Nomura	N/A	N/A
5482859	12/1995	Biller	N/A	N/A
5486475	12/1995	Kramer	N/A	N/A
5518620	12/1995	Eguchi	N/A	N/A
5520812	12/1995	Ryhiner	N/A	N/A
5523003	12/1995	Sell	N/A	N/A
5543039	12/1995	Odegaard	N/A	N/A
5591342	12/1996	Delporte	N/A	N/A
5602719	12/1996	Kinion	N/A	N/A
5647986	12/1996	Nawathe	N/A	N/A
5716689	12/1997	Rogut	N/A	N/A
5725949	12/1997	Pasquali	57/247	B01D 63/033
5762415	12/1997	Tolley	N/A	N/A
5798043	12/1997	Khudenko	N/A	N/A

5910249	12/1998	Kopp	N/A	N/A
5942117	12/1998	Hunter	N/A	N/A
5945002	12/1998	Leukes	N/A	N/A
6001585	12/1998	Gramer	N/A	N/A
6013511	12/1999	Diels	N/A	N/A
6183643	12/2000	Goodley	N/A	N/A
6209855	12/2000	Glassford	N/A	N/A
6214226	12/2000	Kobayashi	N/A	N/A
6241867	12/2000	Mir	N/A	N/A
6299774	12/2000	Ainsworth	N/A	N/A
6309550	12/2000	Iversen	N/A	N/A
6354444	12/2001	Mahendran	N/A	N/A
6361695	12/2001	Husain	N/A	N/A
6367783	12/2001	Raftis	N/A	N/A
6387262	12/2001	Rittmann	N/A	N/A
6387264	12/2001	Baur	N/A	N/A
6485645	12/2001	Husain	N/A	N/A
6531062	12/2002	Whitehill	N/A	N/A
6543753	12/2002	Tharp	N/A	N/A
6555002	12/2002	Garcia	N/A	N/A
6558549	12/2002	Cote	N/A	N/A
6641733	12/2002	Zha	N/A	N/A
6645374	12/2002	Cote	N/A	N/A
6685832	12/2003	Mahendran	N/A	N/A
6692642	12/2003	Josse	N/A	N/A
6706185	12/2003	Goel	N/A	N/A
6743362	12/2003	Porteous	N/A	N/A
6863815	12/2004	Smith	N/A	N/A
6878279	12/2004	Davis	N/A	N/A
6921485	12/2004	Kilian	N/A	N/A
6982036	12/2005	Johnson	N/A	N/A
7169295	12/2006	Husain	N/A	N/A
7186340	12/2006	Rittmann	N/A	N/A
7252765	12/2006	Barnard	N/A	N/A
7294259	12/2006	Cote	N/A	N/A
7300571	12/2006	Cote	N/A	N/A
7303676	12/2006	Husain	N/A	N/A
7318894	12/2007	Juby	N/A	N/A
7622047	12/2008	Koch	N/A	N/A
7699985	12/2009	Cote	N/A	N/A
7713417	12/2009	Sutton	N/A	N/A
7722768	12/2009	Abma	N/A	N/A
8012352	12/2010	Giraldo	N/A	N/A
8545700	12/2012	Stroot	N/A	N/A
8894857	12/2013	Liu	N/A	N/A
2001/0027951	12/2000	Gungerich	N/A	N/A
2002/0158009	12/2001	Khudenko	N/A	N/A
2002/0171172	12/2001	Lowell	N/A	N/A
2003/0092020	12/2002	Carson	N/A	N/A
2003/0104192	12/2002	Hester	N/A	N/A
2003/0173706	12/2002	Rabie	N/A	N/A
2003/0201225	12/2002	Josse	N/A	N/A
2003/0203183	12/2002	Hester	N/A	N/A
2004/0060442	12/2003	Nakahara	N/A	N/A
2004/0065611	12/2003	Jones	N/A	N/A

2004/0079692	12/2003	Cote	N/A	N/A
2004/0115782	12/2003	Paterek	N/A	N/A
2004/0149233	12/2003	Cummins	N/A	N/A
2004/0211723	12/2003	Husain	N/A	N/A
2004/0224396	12/2003	Maston	N/A	N/A
2004/0229343	12/2003	Husain	210/615	C02F 3/1273
2004/0238432	12/2003	Mahendran	N/A	N/A
2004/0251010	12/2003	Doh	N/A	N/A
2005/0051481	12/2004	Husain	N/A	N/A
2005/0064577	12/2004	Berzin	N/A	N/A
2005/0194311	12/2004	Rozich	N/A	N/A
2005/0260739	12/2004	Rosen	N/A	N/A
2005/0269263	12/2004	Rittmann	N/A	N/A
2006/0096918	12/2005	Semmens	N/A	N/A
2006/0124541	12/2005	Logan	N/A	N/A
2006/0163155	12/2005	Chauzy	N/A	N/A
2006/0249449	12/2005	Nakhla	N/A	N/A
2007/0000836	12/2006	Elefritz	N/A	N/A
2007/0012619	12/2006	Thielert	N/A	N/A
2007/0163958	12/2006	Newcombe	N/A	N/A
2007/0235385	12/2006	Barnes	N/A	N/A
2008/0223783	12/2007	Sutton	N/A	N/A
2008/0305539	12/2007	Hickey	N/A	N/A
2009/0095675	12/2008	Runneboom	N/A	N/A
2009/0152762	12/2008	Rave	N/A	N/A
2009/0194477	12/2008	Hashimoto	N/A	N/A
2009/0206026	12/2008	Yoon	N/A	N/A
2010/0012582	12/2009	Frechen	N/A	N/A
2010/0170845	12/2009	Baur	N/A	N/A
2010/0224540	12/2009	Rolchigo	N/A	N/A
2010/0264079	12/2009	Begin	N/A	N/A
2011/0031176	12/2010	Knappe	N/A	N/A
2011/0198283	12/2010	Zha	N/A	N/A
2011/0203992	12/2010	Liu	N/A	N/A
2011/0315629	12/2010	Drogui	N/A	N/A
2012/0000849	12/2011	Fassbender	N/A	N/A
2012/0097604	12/2011	Cote	N/A	N/A
2012/0193287	12/2011	Brouwer	N/A	N/A
2013/0027435	12/2012	Houjou	N/A	N/A
2013/0134089	12/2012	Cote	N/A	N/A
2013/0213883	12/2012	Josse	N/A	N/A
2014/0034573	12/2013	Wenjun	N/A	N/A
2014/0311970	12/2013	Theodoulou	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
2005959	12/1989	CA	N/A
2100002	12/1993	CA	N/A
2102156	12/1993	CA	N/A
2182915	12/1996	CA	N/A
2356316	12/1999	CA	N/A
2300719	12/2000	CA	N/A
2458566	12/2003	CA	N/A
1191768	12/1997	CN	N/A
1569682	12/2004	CN	N/A
1747903	12/2005	CN	N/A

1802322	12/2005	CN	N/A
1830531	12/2005	CN	N/A
101234817	12/2007	CN	N/A
201098607	12/2007	CN	N/A
101316646	12/2007	CN	N/A
101370569	12/2008	CN	N/A
101538101	12/2008	CN	N/A
101790411	12/2009	CN	N/A
101966427	12/2010	CN	N/A
101980969	12/2010	CN	N/A
201850172	12/2010	CN	N/A
102333581	12/2011	CN	N/A
202139109	12/2011	CN	N/A
102665878	12/2011	CN	N/A
102753487	12/2011	CN	N/A
202542950	12/2011	CN	N/A
102826652	12/2011	CN	N/A
202908827	12/2012	CN	N/A
203060938	12/2012	CN	N/A
3544382	12/1986	DE	N/A
3730797	12/1988	DE	N/A
4326603	12/1994	DE	N/A
4440464	12/1995	DE	N/A
10318736	12/2003	DE	N/A
102006034984	12/2007	DE	N/A
0488520	12/1991	EP	N/A
0732141	12/1995	EP	N/A
1496019	12/2004	EP	N/A
0970922	12/2006	EP	N/A
S53135167	12/1977	JP	N/A
S5421057	12/1978	JP	N/A
S5845796	12/1982	JP	N/A
S61120694	12/1985	JP	N/A
H01119397	12/1988	JP	N/A
H02194899	12/1989	JP	N/A
H02207899	12/1989	JP	N/A
H02251299	12/1989	JP	N/A
H03131397	12/1990	JP	N/A
H03249999	12/1990	JP	N/A
H04171096	12/1991	JP	N/A
H0576899	12/1992	JP	N/A
H07148500	12/1994	JP	N/A
H08155498	12/1995	JP	N/A
H08246283	12/1995	JP	N/A
H0985298	12/1996	JP	N/A
H09136100	12/1996	JP	N/A
H10128397	12/1997	JP	N/A
H10337448	12/1997	JP	N/A
H11309480	12/1998	JP	N/A
2000061491	12/1999	JP	N/A
2000070908	12/1999	JP	N/A
2000086214	12/1999	JP	N/A
2002224699	12/2001	JP	N/A
2003053378	12/2002	JP	N/A
2003117590	12/2002	JP	N/A

2003200198	12/2002	JP	N/A
2004290921	12/2003	JP	N/A
2004351324	12/2003	JP	N/A
2005342635	12/2004	JP	N/A
2007050387	12/2006	JP	N/A
2008114215	12/2007	JP	N/A
2008253994	12/2007	JP	N/A
2009285648	12/2008	JP	N/A
20010035160	12/2000	KR	N/A
20050102115	12/2004	KR	N/A
101050375	12/2010	KR	N/A
101179687	12/2011	KR	N/A
20120140329	12/2011	KR	N/A
20130079834	12/2012	KR	N/A
101297685	12/2012	KR	N/A
20130130360	12/2012	KR	N/A
379938	12/1974	SE	N/A
9010488	12/1989	WO	N/A
9426387	12/1993	WO	N/A
0156681	12/2000	WO	N/A
0166474	12/2000	WO	N/A
02094421	12/2001	WO	N/A
2005016498	12/2004	WO	N/A
2008046139	12/2007	WO	N/A
2008130885	12/2007	WO	N/A
2008141413	12/2007	WO	N/A
2009120384	12/2008	WO	N/A
2010094115	12/2009	WO	N/A
2010148517	12/2009	WO	N/A
2011106848	12/2010	WO	N/A
2012019310	12/2011	WO	N/A
2012036935	12/2011	WO	N/A
2012105847	12/2011	WO	N/A
2012145712	12/2011	WO	N/A
2014077888	12/2013	WO	N/A
2014130043	12/2013	WO	N/A
2015142586	12/2014	WO	N/A

OTHER PUBLICATIONS

Semmens et al., “Studies of a Membrane Aerated Bioreactor for Wastewater Treatment,” Membrance Technology, Jul. 1999, vol. 111, pp. 9-13. cited by applicant
Soraunet, “Assessment of Theoretical and Practical Aspects of the Salsnes Filtration Unit,” Civil and Environmental Engineering, Jun. 2012, 90 pages. cited by applicant
Stricker et al, “Pilot Scale Testing of a New Configuration of the Membrane Aerated Biofilm Reactor (MABR) to Treal High-Strength Industrial Sewage,” Water Environ Res, Jan. 2011, vol. 83 (1), pp. 3-14. cited by applicant
Strom, “Technologies to Remove Phosphorus from Wastewater,” Aug. 2006, pp. 1-8. cited by applicant
Sutton et al., “Treating Municipal Wastewater with the Goal of Resource Recovery,” Water Science & Technology, 2011, vol. 63 (1), pp. 25-31. cited by applicant
Syron et al., “Membrane-Aerated Biofilms for High Rate Biotreatment: Performance Appraisal, Engineering Principles, Scale-up, and Development Requirements,” Environmental Science & Technology, Mar. 2008, vol. 42 (6), pp. 1833-1844. cited by applicant
Twarowska-Schmidt et al., “Melt-Spun Asymmetric Poly (4-methyl-1-pentene) Hollow Fibre Membranes,” Journal of Membrane Science, Dec. 1997, vol. 137 (1-2), pp. 55-61. cited by applicant
U.S. Appl. No. 10/777,204, Final Office Action dated Feb. 22, 2006. cited by applicant
U.S. Appl. No. 10/777,204, Non-Final Office Action dated Sep. 26, 2005. cited by applicant

U.S. Appl. No. 10/801,660, Non-Final Office Action dated Dec. 30, 2005. cited by applicant
U.S. Appl. No. 10/895,959, Non-Final Office Action dated Feb. 23, 2007. cited by applicant
U.S. Appl. No. 10/896,086, Non-Final Office Action dated Nov. 1, 2006. cited by applicant
U.S. Appl. No. 11/202,082, Non-Final Office Action dated Mar. 21, 2006. cited by applicant
U.S. Appl. No. 11/203,197, Non-Final Office Action dated Jan. 10, 2007. cited by applicant
U.S. Appl. No. 11/357,051, Non-Final Office Action dated Jan. 16, 2007. cited by applicant
U.S. Appl. No. 11/722,590, Non-Final Office Action dated Apr. 3, 2009. cited by applicant
U.S. Appl. No. 11/722,590, Notice of Allowance dated Sep. 30, 2009. cited by applicant
U.S. Appl. No. 11/949,383, Final Office Action dated Feb. 23, 2009. cited by applicant
U.S. Appl. No. 11/949,383, Non-Final Office Action dated Jun. 12, 2009. cited by applicant
U.S. Appl. No. 11/949,383, Non-Final Office Action dated Sep. 24, 2008. cited by applicant
U.S. Appl. No. 11/949,383, Notice of Allowance dated Jan. 8, 2010. cited by applicant
U.S. Appl. No. 14/769,372, Non-Final Office Action dated Dec. 2, 2016. cited by applicant
U.S. Appl. No. 14/769,461, Advisory Office Action dated May 11, 2020. cited by applicant
U.S. Appl. No. 14/769,461, Final Office Action dated Apr. 29, 2020. cited by applicant
U.S. Appl. No. 14/769,461, Final Office Action dated Mar. 14, 2019. cited by applicant
U.S. Appl. No. 14/769,461, Non-Final Office Action dated Mar. 2, 2018. cited by applicant
U.S. Appl. No. 14/769,461, Non-Final Office Action dated Nov. 5, 2019. cited by applicant
U.S. Appl. No. 14/769,461, Notice of Allowance dated May 22, 2020. cited by applicant
U.S. Appl. No. 14/769,461, Restriction Requirement dated Dec. 18, 2017. cited by applicant
U.S. Appl. No. 16/178,974, Non Final Office Action dated Jul. 27, 2022. cited by applicant
U.S. Appl. No. 16/178,974, Advisory Action dated Dec. 3, 2021. cited by applicant
U.S. Appl. No. 16/178,974, Advisory Action dated Jan. 9, 2023. cited by applicant
U.S. Appl. No. 16/178,974, Advisory Action dated Jun. 9, 2022. cited by applicant
U.S. Appl. No. 16/178,974, Final Office action dated Apr. 5, 2022. cited by applicant
U.S. Appl. No. 16/178,974, Final Office Action dated Nov. 1, 2022. cited by applicant
U.S. Appl. No. 16/178,974, Final Office Action dated Oct. 22, 2020. cited by applicant
U.S. Appl. No. 16/178,974, Final office Action dated Oct. 7, 2021. cited by applicant
U.S. Appl. No. 16/178,974, Non-Final Office Action dated Dec. 21, 2021. cited by applicant
U.S. Appl. No. 16/178,974, Non-Final Office Action dated Jan. 14, 2020. cited by applicant
U.S. Appl. No. 16/178,974, Non-Final Office Action dated Jun. 25, 2021. cited by applicant
U.S. Appl. No. 16/178,974, Non-Final Office Action dated Jun. 5, 2020. cited by applicant
U.S. Environmental Protection Agency, “Municipal Nutrient Removal Technologies Reference Document”, cPA 832-R-08-006, Sep. 2008, 449 pages. cited by applicant
Wang et al., “Nitritation Performance and Biofilm Development of CO-and Counter-Diffusion Biofilm Reactors Modelling and Experimental Comparison,” Water Res, Jun. 2009, vol. 43 (10), pp. 2699-2709. cited by applicant
Wei et al., “Mixed Pharmaceutical Wastewater Treatment by Integrated Membrane-Aerated Biofilm Reactor (MABR) System—A Pilot-scale Study,” Bioresource Technology, 2012, vol. 122, pp. 189-195. cited by applicant
Woolard, “The Advantages of Periodically Operated Biofilm Reactors for the Treatment of Highly Variable Wastewater,” Water Science and Technology, Dec. 1997, vol. 35 (1), pp. 199-206. cited by applicant
Xin Wei et al., “Mixed pharmaceutical wastewater treatment by integrated membrane-aerated biofilm reactor (MABR) system—A pilot-scale study,” Bioresource Technology, vol. 122, pp. 189-195 (Jun. 29, 2012). cited by applicant
Xing et al., “Microfiltration-Membrane-Coupled Bioreactor for Urban Wastewater Reclamation,” Desalination, Dec. 2001, vol. 141 (1), pp. 63-73. cited by applicant
Yamagiwa et al., “Simultaneous Organic Carbon Removal and Nitrification by Biofilm Formed on Oxygen Enrichment Membrane,” Journal of Chemical Engineering of Japan, Oct. 1994, pp. 638-643. cited by applicant
Yaoliang et al., “New Technologies for Biological Wastewater Treatment: Theory and Application,” China Environmental Science Press, Version 2, p. 338. cited by applicant
Yeh et al., “Pure Oxygen Fixed Film Reactor,” Journal of the Environmental Engineering Division, Aug. 1978, vol. 104 (4), pp. 611-623. cited by applicant
Ferrero, “Development of an Air-Scour Control System for Membrane Bioreactors,” Universitat de Girona, 2011, 148 pages. cited by applicant
Gerber et al., “Interactions Between Phosphate, Nitrate and Organic Substrate in Biological Nutrient Removal Processes,” Water Science and Technology, Jan. 1987, vol. 19 (1-2), pp. 183-194. cited by applicant

Indian Patent Application No. 201647035519, Office Action dated Aug. 28, 2019. cited by applicant
Indian Patent Application No. 202048006620, Office Action dated Dec. 17, 2021. cited by applicant
Indian Patent Application No. 7531/DELNP/2008, Office Action dated Jul. 19, 2013. cited by applicant
International Patent Application No. PCT/CA2004/000206, International Preliminary Report on Patentability and Written Opinion dated Aug. 19, 2005. cited by applicant
International Patent Application No. PCT/CA2004/000206, International Search Report dated May 19, 2004. cited by applicant
International Patent Application No. PCT/CA2004/001495, International Preliminary Report on Patentability and Written Opinion dated Feb. 21, 2006. cited by applicant
International Patent Application No. PCT/CA2004/001495, Written Opinion dated Feb. 1, 2005. cited by applicant
International Patent Application No. PCT/CA2004/001496, International Preliminary Report on Patentability and Writter Opinion dated Feb. 21, 2006. cited by applicant
International Patent Application No. PCT/CA2004/001496, Written Opinion dated Jan. 7, 2005. cited by applicant
International Patent Application No. PCT/CA2005/001250, International Preliminary Report on Patentability and Written Opinion dated Feb. 13, 2007. cited by applicant
International Patent Application No. PCT/CA2005/001250, Written Opinion dated Dec. 8, 2005. cited by applicant
International Patent Application No. PCT/US2013/027403, International Preliminary Report on Patentability dated Sep. 3, 2015. cited by applicant
International Patent Application No. PCT/US2013/027403, International Search Report dated Oct. 1, 2013. cited by applicant
International Patent Application No. PCT/US2013/027411, International Preliminary Report on Patentability dated Sep. 3, 2015. cited by applicant
International Patent Application No. PCT/US2013/027411, International Search Report dated Nov. 7, 2013. cited by applicant
International Patent Application No. PCT/US2013/027435, International Preliminary Search Report dated Sep. 3, 2015. cited by applicant
International Patent Application No. PCT/US2013/027435, International Search Report dated Sep. 9, 2013. cited by applicant
International Patent Application No. PCT/US2014/031321, International Preliminary Report on Patentability dated Sep. 29, 2016. cited by applicant
International Patent Application No. PCT/US2014/031321, International Search Report dated and Written Opinion dated Dec. 19, 2014. cited by applicant
International Patent Application No. PCT/US2015/019943, International Preliminary Reporton Patentability dated May 17, 2018. cited by applicant
International Patent Application No. PCT/US2015/019943, International Search Report and Written Opinion dated Mar. 21, 2018. cited by applicant
Israel Patent Application No. 247625, Office Action dated Aug. 29, 2019. cited by applicant
Israel Patent Application No. 247625, Office Action dated Dec. 22, 2020. cited by applicant
Joss et al., "Combined Nitritation-Anammox: Advances in Understanding Process Stability," Environmental Science & Technology, Nov. 2011, vol. 45 (22), pp. 9735-9742. cited by applicant
Korean Patent Application No. 10-2015-7025449, Intellectual Property Trial and Appeal Board Decision dated Jan. 21, 2021. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Feb. 17, 2020. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Feb. 9, 2023. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Jan. 31, 2019. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Jul. 20, 2021. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Oct. 30, 2019. cited by applicant
Korean Patent Application No. 10-2015-7025449, Office Action dated Oct. 6, 2022. cited by applicant
Korean Patent Application No. 10-2016-7028984, Office Action dated Apr. 28, 2022. cited by applicant
Korean Patent Application No. 10-2016-7028984, Office Action dated Feb. 3, 2022. cited by applicant
Korean Patent Application No. 10-2016-7028984, Office Action dated Jul. 28, 2021. cited by applicant
Korean Patent Application No. 10-2020-7002799, Office Action dated Feb. 18, 2020. cited by applicant
Korean Patent Application No. 10-2020-7002799, Office Action dated Jan. 28, 2021. cited by applicant
Korean Patent Application No. 10-2020-7002799, Office Action dated Mar. 19, 2021. cited by applicant
Korean Patent Application No. 10-2020-7008099, Office Action dated Feb. 9, 2023. cited by applicant

Korean Patent Application No. 10-2020-7008099, Office Action dated Jan. 25, 2021. cited by applicant
Korean Patent Application No. 10-2022-7033431, Office Action dated Oct. 28, 2022. cited by applicant
Li et al., "Membrane Aerated Biofilm Reactors: A Brief Current Review," Recent Patents on Biotechnology, 2008, vol. 2, pp. 88-93. cited by applicant
Liang, et al., "Electronic Technology Training Course," China Electric Power Press, 2009, pp. 189. cited by applicant
Lindeke et al., "The Role and Production of VFAs in a Highly Flexible BNR Plant," WEFTEC, Jan. 2005, pp. 1151-1173. cited by applicant
Ljunggren, "Micro Screening in Wastewater Treatment—An Overview," Vatten, 2006, vol. 62, pp. 171-177. cited by applicant
Marcelo et al., "The Air-Based Membrane Biofilm Reactor (MBFR) for Energy Efficient Wastewater Reatment", WEFTEC 2012: Session 71 through Session 80, pp. 5458-5485. cited by applicant
Martin et al., "The Membrane Biofilm Reactor (MBfR) for Water and Wastewater Treatment: Principles, Applications, and Recent Developments," Bioresource Technology, Oct. 2012, vol. 122, pp. 83-94. cited by applicant
Narayanan et al, "Fermentation of Return Activated Sludge to Enhance Biological Phosphorus Removal." WEFTEC, 2002, 7 pages. cited by applicant
Semmens et al., "COD and Nitrogen Removal by Biofilms Growing on Gas Permeable Membranes," Water Research, Elsevier, Nov. 1, 2003, vol. 37(18), pp. 4343-4350. cited by applicant
Korean Patent Application No. 10-2020-7008099, Office Action dated Nov. 14, 2023. cited by applicant
Korean Patent Application No. 10-2020-7008099, Office Action dated Mar. 25, 2024. cited by applicant
"Drainage Facility Standards," Korea Water and Wastewater Association, 2011, pp. 298, 378, 394, 395, 531. cited by applicant
Australian Patent Application No. 2019203733, Office Action dated Jul. 16, 2020. cited by applicant
Australian Patent Application No. 2019203733, Office Action dated Mar. 20, 2020. cited by applicant
Australian Patent Application No. AU2013378841, Office Action dated Sep. 14, 2017. cited by applicant
Barajas et al., "Fermentation of a Low VFA Wastewater in an Activated Primary Tank," Water SA, Jan. 2002, vol. 28 (1), pp. 89-98. cited by applicant
Barnard et al., "Using Alternative Parameters to Predict Success for Phosphorus Removal in WWTP's," WEFTEC, 2005, pp. 1970-1984. cited by applicant
Baur et al., "Primary Sludge Fermentation—Results From Two Full-Scale Pilots at South Austin Regional (TX, USA) and Durham AWWTP (OR, USA)," WEFTEC, 2002, 23 pages. cited by applicant
Brindle et al., "Nitrification and Oxygen Utilisation in a Membrane Aeration Bioreactor," Journal of Membrane Science, Jun. 1998, vol. 144 (1-2), pp. 197-209. cited by applicant
Canadian Patent Application No. 2,901,764, Office Action dated Jul. 12, 2019. cited by applicant
Canadian Patent Application No. 2,901,764, Office Action dated Nov. 22, 2018. cited by applicant
Canadian Patent Application No. 2,943,072, Office Action dated Feb. 1, 2022. cited by applicant
Canadian Patent Application No. 2,943,072, Office Action dated Sep. 8, 2022. cited by applicant
Casey et al., "Review of Membrane Aerated Biofilm Reactors," Resources, Conservation and Recycling, Jul. 1999, vol. 27 (1-2), pp. 203-215. cited by applicant
Chen, Guanwen et al., "Recent Advances in Membrane Technology and Engineering Applications", National Defense Industry Press, Aug. 2013, pp. 164-165. cited by applicant
Chinese Patent Application No. 200480004060.5, Office Action dated Dec. 15, 2006. cited by applicant
Chinese Patent Application No. 201380073639.6, Office Action dated Apr. 25, 2016. cited by applicant
Chinese Patent Application No. 201380073677.1, Office Action dated Jul. 21, 2016. cited by applicant
Chinese Patent Application No. 201380073677.1, Office Action dated Mar. 10, 2017. cited by applicant
Chinese Patent Application No. 201380073696, Office Action dated Feb. 25, 2019. cited by applicant
Chinese Patent Application No. 201380073696, Office Action dated Jul. 25, 2019. cited by applicant
Chinese Patent Application No. 201380073696.4, Office Action dated Jun. 12, 2016. cited by applicant
Chinese Patent Application No. 201380073696.4, Reexamination Decision dated Jan. 16, 2020. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action and Search Report dated Jul. 19, 2022. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action dated Aug. 31, 2020. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action dated Jan. 30, 2022. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action dated Jan. 6, 2020. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action dated Mar. 23, 2021. cited by applicant
Chinese Patent Application No. 201580026027.0, Office Action dated Nov. 3, 2022. cited by applicant

Chinese Patent Application No. 202010227987.6, Office Action dated Jul. 18, 2022. cited by applicant
Chinese Patent Application No. 202010227987.6, Office Action dated Mar. 7, 2022. cited by applicant
Chinese Patent Application No. 202010227987.6, Office Action dated Sep. 3, 2021. cited by applicant
Chinese Patent Application No. 202010300259.3, Office Action dated May 16, 2022. cited by applicant
Chinese Patent Application No. 202010300259.3, Office Action dated Oct. 11, 2021. cited by applicant
Chinese Patent Application No. 202010300259.3, Office Action dated Oct. 21, 2023. cited by applicant
Chinese Patent Application No. CN201380073696.4, Office Action dated Nov. 6, 2017. cited by applicant
Cote et al., “Bubble-free aeration using Membranes: Mass Transfer Analysis,” Journal of Membrane Science, Nov. 1989, vol. 47 (1-2), pp. 91-106. cited by applicant
Cote et al., “Bubble-Free Aeration Using Membranes: Process Analysis,” Journal Water Pollution Control Federation, Nov. 1998, vol. 60 (11), pp. 1986-1992. cited by applicant
Cui et al., “Innovative Application of Carbon Tube Aerated Membranes to Enhance Anaerobic Baffled Reactor for Wastewater Treatment,” Environmental Science, May 2008, vol. 29(5), pp. 1216-1220. cited by applicant
Diamond et al., “Model of Sustainability,” Water & Wastes Digest, Sep. 2013, pp. 34-35, [retrieved on Jul. 3, 2015], Retrieved from the Internet:
[URL:https://www.gewater.com/kcpguest/documents/TechnicalPapers_Cust/Americas/English/WaterWastesDigest_T
. . . . cited by applicant
Downing et al, “Nitrogen Removal from Wastewater Using a Hybrid Membrane-Biofilm Process:Pilot-Scale Studies,” Water Environment Research, Mar. 2010, vol. 82 (3), pp. 195-201. cited by applicant
Downing et al., “Effect of Bulk Liquid BOD Concentration on Activity and Microbial Community Structure of a Nitrifying, Membrane-Aerated Biofilm,” Applied Microbiology and Biotechnology, Nov. 2008, vol. 81 (1), pp. 153-162. cited by applicant
Envities, Lamella Sedimentation Tanks and Clarifiers, Nov. 21, 2010, pp. 1-4. cited by applicant
European Application No. 20164903.5, Communication pursuant to Rule 69 EPC, dated Aug. 31, 2020. cited by applicant
European Patent Application No. 05774824.6, Office Action dated Apr. 16, 2008. cited by applicant
European Patent Application No. 05774824.6, Office Action dated Aug. 13, 2009. cited by applicant
European Patent Application No. 05774824.6, Supplementary Partial European Search Report dated Jan. 25, 2008. cited by applicant
European Patent Application No. 13709632.7, Communication pursuant to Article 94(3) EPC dated Jun. 27, 2017. cited by applicant
European Patent Application No. 15713275.4, Communication pursuant to Article 94(3) EPC dated Apr. 3, 2020. cited by applicant
European Patent Application No. 20164903.5, European Communication dated Mar. 15, 2023. cited by applicant
European Patent Application No. 20164903.5, Extended European Search Report dated Jul. 20, 2020. cited by applicant

---

Primary Examiner: Huang; Ryan B

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 16/998,704, filed Aug. 20, 2020, which is a of continuation of U.S. application Ser. No. 14/769,461, filed Aug. 21, 2015, issued on Sep. 22, 2020 as U.S. Pat. No. 10,781,119, which is a National Stage Entry of International Application No. PCT/US2013/027435, filed Feb. 22, 2013. U.S. application Ser. Nos. 16/998,704 and 14/769,461 and International Application No. PCT/US2013/027435 are incorporated herein by reference.

FIELD
(1) This specification relates to wastewater treatment, to membrane biofilm reactors, and to assemblies of gas permeable membranes for supporting a biofilm.
BACKGROUND
(2) In a membrane biofilm reactor (MBfR), a membrane is used to both support a biofilm and to transport a gas to the biofilm. Membrane biofilm reactors were recently reviewed by Martin and Nerenberg in “The

membrane biofilm reactor (MBfR) for water and wastewater treatment: Principles, applications, and recent developments" (Bioresour. Technol. 2012). Membrane-aerated biofilm reactors (MABR) are a subset of MBfRs in which an oxygen containing gas is used. MABRs were reviewed by Syron and Casey in "Membrane-Aerated Biofilms for High Rate Biotreatment: Performance Appraisal, Engineering Principles, Scale-up, and Development Requirements" (Environmental Science and Technology, 42(6): 1833-1844, 2008).
(3) U.S. Pat. No. 7,169,295 describes a membrane supported biofilm reactor with modules having fine hollow fiber membranes. The membranes are made from dense wall polymethyl pentene (PMP) used in tows or formed into a fabric. The membranes are potted in a header of a module to enable oxygen containing gas to be supplied to the lumens of the hollow fibers. The reactor may be used to treat wastewater. Mechanical, chemical and biological methods are used to control the thickness of the biofilm.
SUMMARY OF THE INVENTION
(4) This specification describes an assembly, alternatively called a cord, which may be used for supporting a biofilm. The cord comprises a plurality of hollow fiber gas transfer membranes. The cord may optionally also comprise one or more reinforcing filaments.
(5) The cord may comprise a plurality of yarns. At least one of the yarns comprises a plurality of gas transfer membranes. At least one of the yarns extends along the length of the cord generally in the shape of a spiral. In some cases, the cord has a core and one or more wrap yarns. In some other cases, the cord comprises a set of braided yarns.
(6) The cord preferably has an outside diameter in the range of about 0.3 mm to 2.0 mm. The gas transfer membranes preferably have an outside diameter that is less than 200 microns. The sum of the circumferences of the gas transfer membranes is preferably at least 1.5 times the circumference of the smallest circle that can surround the cord. In use, a biofilm covers the cord and the outer surface of the biofilm is substantially round.
(7) A module may be made by potting a plurality of cords in at least one header. The cords are generally independent of each other except in the header. A reactor may be made by placing the module in a tank adapted to hold water to be treated and providing a gas delivery system. A process for treating wastewater comprises steps of feeding water to the tank and supplying a gas to the module. In use, a biofilm may cover a cord to form a membrane biofilm assembly.
(8) The cord, module, reactor and process may be used to treat water, for example in, or in the manner of, an MBfR.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) FIG. **1** is a photograph of a cord.
(2) FIGS. **2** to **4** are photographs of alternative cords.
(3) FIG. **5** is a schematic drawing of a machine for making a cord.
(4) FIGS. **6** to **8** are schematic drawings of steps in a process for making a module comprising a plurality of cords.
(5) FIG. **9** is a schematic drawing of a module comprising a plurality of cords.
(6) FIG. **10** is a schematic drawing of a reactor comprising the module of FIG. **9**.
(7) FIGS. **11** to **14** are drawings of the cords in FIGS. **1** to **4** respectively.
DETAILED DESCRIPTION
(8) FIGS. **1** to **4** and FIGS. **11** to **14** each show a cord **10** comprising a plurality of yarns **8**. At least one of the yarns **8** comprises a plurality of hollow fiber gas transfer membranes **14**. Preferably, at least one of the yarns **11** comprises at least one reinforcing filament **34**.
(9) At least one of the yarns **8** extends along the length of the cord **10** generally in the shape of a spiral and may be referred to as a spiral yarn. Preferably, at least one spiral yarn **8** is wrapped at least partially around the outside of the other yarn or yarns **8** of the cord **10**.
(10) In the cords **10** of FIGS. **1** to **4** and FIGS. **11** to **14**, the yarns **8** are arranged to provide a core **12** and one or more wraps **18**. A wrap **18** travels in a spiral that is always outside of another yarn **8**. For example, a wrap **18** may spiral around a core **12**, or around one or more other wraps **18** that spiral around the core **12**. An only or outer wrap **18** is located entirely outside of the other yarn or yarns **8** of a cord **10**. Alternatively, a spiral yarn **8** may be wrapped only partially outside of the other yarns or yarns **8** or a cord. For example, two or more spiral yarns **8** may be twisted around each other or four or more spiral yarns **8** may be braided together

to form a cord **10** with no wrap **18**. In another alternative, four or more spiral yarns **8** may be braided together around a core **12**.
(11) Each cord **10** has a plurality of hollow fiber gas transfer membranes **14**. The gas transfer membranes **14** may be located in a core **12**, in one or more wraps **18**, or in another yarn **8**. The gas transfer membranes **14** are preferably provided in the form of a multi-filament yarn having a plurality of gas transfer membranes which may be called a gas transfer membrane yarn **15**. Optionally, a cord may also have a reinforcing yarn **16**. The reinforcing yarn has one or more reinforcing filaments **34**. A yarn **8** having both a plurality of gas transfer membranes **14** and at least one reinforcing filament **34** may be called both a gas transfer membrane yarn **15** and a reinforcing yarn **16**.
(12) The outside diameter of a cord **10** is preferably in a range of about 0.3 to 2.0 mm. The outside diameter of the cord **10** may be measured as the largest width of a cord measured through its longitudinal axis or as the diameter of the smallest hole that the cord **10** will pass through. Anomalies, defects or non-repeating bumps are ignored in these measurements.
(13) Generally speaking, a core **12** provides mechanical strength, defines a longitudinal axis of the cord **10**, supports any wraps **18**, and may also comprise gas transfer membranes **14**. A wrap **18**, or other yarn outside of the core **12**, may do one or more of: protect the core **12** or another underlying yarn **8**, comprise gas transfer membranes **14**, or contribute to the mechanical strength of the cord **10**.
(14) A core **12** can be made of one or more monofilament yarns or multi-filament yarns. A Multi-filament yarn may comprise filaments that are braided, twisted or otherwise united, or filaments that are merely collected together in a bundle or tow. Multiple yarns may be arranged as parallel warp yarns or twisted, braided or otherwise united. For example a core **12** may consist essentially of a single monofilament yarn; a single multi-filament yarn; or, an assembly of about 2 to 6 monofilament or multi-filament yarns arranged in parallel or twisted or braided together. An assembly of twisted or braided yarns may be preferred since the assembly will be more flexible than a single monofilament of the same outer diameter. A core **12** typically, but not necessarily, comprises at least one reinforcing yarn **16**. A core **12** may optionally comprise one or more gas transfer membrane yarns **15**.
(15) A wrap **18** is typically a multi-filament yarn. A multi-filament yarn may comprise filaments that are twisted or otherwise united or filaments that are merely collected together in a bundle or tow. A wrap **18** can be wrapped around a core **12** in a clockwise spiral or a counterclockwise spiral. Alternatively, a cord **10** may have at least one wrap **18** in each direction or no wrap **18**. A wrap **18** can be a gas transfer membrane yarn **15**, a reinforcing yarn **16**, or both.
(16) A reinforcing filament **34** can be made of any water-resistant and non-biodegradable polymer such as polyethylene, nylon or polyester, preferably nylon or polyester. A reinforcing filament **34** is typically solid. Gas transfer membranes **14** tend to be expensive and weak relative to reinforcing filaments **34** made of common textile polymers such as nylon or polyester. A reinforcing yarn **16** can be a monofilament or multi-filament yarn. In the case of a multi-filament yarn, the reinforcing filaments **34** may be braided, twisted or otherwise united, or filaments that are merely collected together in a bundle or tow. Optionally, a yarn **8** may comprise one or more reinforcing filaments **34** mixed with the gas transfer membranes **14**.
(17) The gas transfer membranes **14** preferably have an outside diameter of 500 microns or less, more preferably 200 microns or less, optionally 100 microns or less. The hollow area of a hollow fiber (meaning the cross sectional area of the lumen of a fiber as a percentage of its total cross sectional area) is preferably at least 20%, for example in the range of 20-50%. For example, a gas transfer membrane **14** may have an outside diameter in the range of about 30-70 microns and an inside diameter of about 15-50 microns. The wall thickness of a gas transfer membrane **14** may be 20 microns or less.
(18) The gas transfer membranes **14** are preferably handled in a multi-filament gas transfer membrane yarn **15**. A gas transfer membrane yarn **15** may have between 2 and 200, between 12 and 96, or between 10 and 60 individual filaments of gas transfer membrane **14**. A gas transfer membrane yarn **15** used as a wrap **18** is preferably not tightly twisted, braided or crimped to allow the individual gas transfer membranes **14** to spread out over an underlying yarn **8**. A gas transfer membrane yarn **15** may be made by re-winding gas transfer membranes **14** from multiple take up spools in combination onto another spool.
(19) A gas transfer membrane yarn **15** may be provided, in a core **12**, either as a central yarn or as a warp parallel to a central reinforcing yarn **16**; in one or more wraps **18**; or, in another spiral yarn **8**. It is desirable for gas transfer efficiency to have the gas transfer membranes **14** near the outer surface of the cord **10**. However, the gas transfer membranes **14** are typically fragile and they are more likely can be damaged if they form the outer surface of a cord **10**. Accordingly, it is preferable for a reinforcing yarn **16** to be used as an outer wrap

18. In this case, a gas transfer membrane yarn **15** may be used in the core **12** or in an inner wrap **18**.
(20) The gas transfer membranes **14** may be porous, non-porous or semi-porous. Composite membranes, for example having a non-porous membrane layer, and a semi-porous or porous support layer, may also be used. Asymmetric membranes, for example having a non-porous region and an integral semi-porous or porous region, may also be used.
(21) Porous gas transfer membranes **14** may have pores up to the microfiltration range. Wetting is avoided by choosing hydrophobic materials or treating the hollow fibers **14** to make them hydrophobic. Porous hollow fibers **14** may be made, for example, using polyethylene, polyvinylchloride, polypropylene or polysulfone.
(22) Non-porous gas transfer membranes **14**, including dense wall gas transfer membranes **14**, may be made from a thermoplastic polymer, for example a polyolefin such as polymethyl pentene (Poly (4-methylpentene-1) or PMP), polyethylene (PE) or polypropylene (PP). PMP is sold, for example, by Mitsui Petrochemical under the trade mark TPX. The polymer may be melt spun into a hollow fiber. The gas transfer membranes **14** may be called non-porous if water does not flow through the fiber walls by bulk or advective flow of liquid even though there are small openings through the wall, typically in the range of 4 or 5 Angstroms in the case of melt spun PMP. However, oxygen or other gases may permeate or travel through the fiber walls. In a dense walled hollow fiber **14**, gas travel is primarily by molecular diffusion or dissolution-diffusion which occurs when openings in the fiber walls are generally less than 30 Angstroms. Gas transfer membranes **14** as described in U.S. Pat. No. 7,169,295, which is incorporated by reference, may be used.
(23) The term porous has been used to refer to any structure having openings larger than in a dense wall, for example having openings of 30 or 40 Angstroms or more, but without openings large enough to be wetted or transport liquid water by advective, Poiseuille or bulk flow. In this specification, membranes with openings in this size range are referred to as semi-porous.
(24) Gas transfer membranes **14** may alternatively be made by mechanically or thermally treating a melt spun thermoplastic polymer after spinning to increase its permeability to oxygen without making the fiber wettable or capable of permitting advective flow of liquid water. Spinning or post-treatment steps that can be used or controlled to increase permeability include the spinning speed or drawing ratio, the quenching conditions such as temperature or air flow rate, post annealing, if any, stretching and heat setting. The resulting fibers may have a dense layer, with openings ranging from the size of openings in the raw polymer to 30 or 40 Angstroms, on either the inside of the fiber, the outside of the fiber or both, with the remaining parts of the fiber being porous or semi-porous. For example, U.S. Pat. No. 4,664,681, issued on May 12, 1997, to Anazawa et al. describes, in examples 4 and 6, processes for melt-spinning and post-processing PE and PP to produce acceptable fibres while other fibers are made from PMP or polyoxymethylene. Processes described in "Melt-spun Asymmetric Poly (4-methyl-1-pentene) Hollow Fibre Membranes", Journal of Membrane Science, 137 (1997) 55-61, Twarowska-Shmidt et al., also produce acceptable fibres of PMP and may be adopted to produce fibres of other polyolefins such as PE or PP. In one example, the mean pore size of the fibers produced is just over 40 Angstroms. In U.S. Pat. No. 4,664,681, membranes are melt spun, stretched (by producing the membrane at a high draft ratio) under weak cooling and then heat treated. The resulting membranes are asymmetric containing a dense layer with substantially no pore with a diameter of 30 Angstroms or more and a microporous layer having larger pores. The non-porous layer is the outer surface of the fiber and so the fiber is non-wetting.
(25) Another alternative process for making gas transfer membranes **14** is to make an asymmetric outside dense skin membrane with a spongy substructure by the non-solvent induced phase separation (NIPS) process. Polymers typically used for this process are polysulfone, cellulose acetate and polyimide. Other alternative methods of making gas transfer membranes **14** may include, for example, meltblown extrusion, flash spinning, and electrospinning.
(26) In general, silicon rubber or PDMS have very high oxygen permeability but cannot be processed using many textile techniques and are not available in small diameter fibers. PMP has higher oxygen permeability than PE or PP, but it is more expensive. Porous membranes have high oxygen permeability but they are prone to wetting in use. Accordingly, dense wall polymeric gas transfer membranes **14**, for example of PE. PP or PMP, with a wall thickness of 50 microns or less, preferably 20 microns or less, or polymeric membranes that are not entirely dense walled but have a nonporous or semi-porous layer, are preferred but not essential.
(27) A yarn **8** in a core **12**, whether it is a gas transfer yarn **15** or a reinforcing yarn **16**, may be called a warp **26**. A wrap **18** is preferably applied around a core **12** or another underlying yarn **8** with some tension to cause its filaments to spread on the surface of the core **12**. Wrapping may be done with a pitch ratio (pitch divided by the diameter of the core) of between 1 and 5. Wrapping can be in one direction only, but is preferably done

in both directions. There may be 1 or more, for example between 1 and 3, wraps **18** in one direction. Gas transfer membranes **14** are preferably directly exposed over at least 25% of the surface of the core **12** although oxygen can also travel from a gas transfer membrane **14** through an overlying yarn **8**.

(28) A cord **10** may consist of only a set of twisted yarns **8**. However, merely twisted yarns **8** may tend to partially untwist and separate in use. Accordingly, it is preferable to use twisted yarns **8** as a core **12** with at least one wrap **18** wrapped around the core **12** in the direction opposite to the twist of the core **12**. A braided core **12** is more stable. Spiral yarns **8** may be added as a braid around a core **12**, but a wrap **18** can be made at a faster line speed with a less complicated machine. A cord structure using one or more wraps **18** also allows for a reinforcing yarn **16** to be used as an outer wrap **18** to help protect an interior gas transfer membrane yarn **15**.

(29) In the examples of FIGS. **1** to **4** and FIGS. **11** to **14**, gas transfer membrane yarns **15** are made up of 48 dense wall PMP hollow fiber filaments. The filaments have an outside diameter of less than about 70 microns and a wall thickness of less than 20 microns. Reinforcing yarns **16** are either monofilament yarns or multi-filament yarns of polyester (PET).

(30) In FIG. **1** and FIG. **11**, a first cord **10***a* has a core **12** made up of a single monofilament reinforcing yarn **16** and a two multi-filament gas transfer membrane yarns **15** applied as warps **26** parallel to the reinforcing yarn **16**. The core **12** is covered with two wraps **18**, one in each direction, with a wrapping pitch of 1.8. Each wrap **18** is a single multi-filament gas transfer membrane yarn **15**.

(31) In FIG. **2** and FIG. **12**, a second cord **10***b* has a core **12** comprising a reinforcing yarn **16** and a gas transfer membrane yarn **15**. Each of these yarns in the core **12** is an untwisted multi-filament yarn, alternatively called a tow. The second cord **10***b* also has two wraps **18**, one in each direction. Each wrap **18** is an untwisted multi-filament reinforcing yarn **16**.

(32) In FIG. **3** and FIG. **13**, a third cord **10***c* has a core **12** comprising a set of multifilament reinforcing yarns **16** braided together. The third cord **10***c* also has an inner wrap **18** comprising an untwisted multifilament gas transfer membrane yarn **15** and an outer wrap comprising an untwisted multifilament reinforcing yarn **16**.

(33) In FIG. **4** and FIG. **14**, a fourth cord **10***c* has a core **12** comprising a set of multifilament reinforcing yarns **16** braided together and an untwisted multifilament gas transfer membrane yarn **15** as a warp **26** parallel to the reinforcing yarns **16**. The gas transfer membrane yarn **15** is parallel with the core **12** but not braided with the reinforcing yarns **16**. The third cord **10***c* also has a wrap **18** comprising an untwisted multifilament reinforcing yarn **16**.

(34) FIG. **5** shows a machine **20** for making a cord **10**. The machine **20** is built on frame **22** that supports the different components and aligns them. One or more warps **26** are supplied to the machine **20** from a creel **24**. The creel **24** has stationary bobbin holders, guides and tensioning devices, as found in other textile equipment. The warps **26** pass through a distributor **28**. The distributor **28** may have a central opening and one or more eyelets around the central opening. A warp **26** is unwound from a bobbin on the creel **24**, positioned to the top of the distributor **28** through a roller and fed vertically down through the distributor **28**. A take up winder (not shown) pulls the cord **10** downwards through the machine **20** and onto a bobbin. The one or more warps **26** form the core **12** of the cord **10**.

(35) One or more spindles **30**, or other yarn wrapping devices, are located below the distributor **28**. Each spindle **30** is loaded with a yarn and wraps the yarn around the one or more warps **26** of the core **12** as they pass through the spindle **30**. Due to the downward movement of the core **12**, each wrapped yarn forms a spiral wrap **18**. The machine **20** may also have alignment guides (not shown) to keep the core **12** aligned with the central axis of the spindles **30** and to reduce vibration of the core **12**.

(36) An example of a suitable spindle **30** is a Temco™ spindle model MSE150 by Oerlikon Textile. Each spindle **30** has an electrical motor and a hollow core and holds a bobbin of wrap yarn. The spindle **30** is positioned so that its central axis coincides with the core **12**. In the machine **20** of FIG. **2**, there are two spindles **30**, one rotating clockwise and the other rotating counter clockwise. The spindles **30** can rotate at an adjustable speed of up to 25,000 rpm to provide a controllable pitch. Alternatively, a rotating creel may be used in place of the spindle **30**. In a rotating creel, bobbins are mounted on a wheel that rotates in one direction around the core **12** without being in contact with it. Each bobbin is preferably equipped with tension control.

(37) A plurality of cords **10**, for example 100 or more, may be made into a module generally in the manner of making an immersed hollow fiber membrane filtration module. At least one end of each of the cords **10** is potted in a block of a potting material such as thermoplastic or thermosetting resin which is sealed to a pan to form a header. The ends of the gas transfer membranes **14** are made open to the inside of the header, for

example by cutting them open after potting. The other ends of the cords **10** may be potted in another header with the ends of the gas transfer membranes **14** open or closed, closed individually, or looped back and potted in the first header. A port in the header allows a gas to be fed to the lumens of the gas transfer membranes **14**. The gas may be fed to the gas transfer membranes **14** in a dead end manner or with exhaust through a second header.

(38) As one example, the composite fibers **10** may be assembled into modules and cassettes according to the configuration of ZeeWeed 500™ immersed membrane filtration units sold by GE Water & Process Technologies. Sheets of cords **10** are prepared with the composite fibers **10** generally evenly spaced in the sheet. Multiple sheets are stacked on top of each other to form a bundle with adjacent sheets spaced apart from each other. The bundle is potted. After the potting material cures, it is cut to expose the open ends of the gas transfer membranes and sealed to a header pan. Several such modules may be attached to a common frame with their ports manifolded together to form a cassette. Various useful techniques that may be used or adapted for making a module are described in U.S. Pat. Nos. 7,169,295, 7,300,571, 7,303,676, US Publication 2003/01737006 A1 and International Publication Number WO 02/094421, all of which are incorporated by reference. Alternatively, other known techniques for making a hollow fiber membrane module may be used.

(39) Referring to FIG. **6**, multiple cords **10**, or an undulating cord **10**, are laid out on a flat jig or drum to provide a set of generally parallel segments of cord **10** in a sheet **38**. The segments of cord **10** may be kept evenly spaced from each other in the sheet **38**, for example by a woven filament **40** or a strip of hot melt adhesive **42**. When segments of cord **10** are used, the ends of the cords **10** may be sealed, for example by melting them with an iron or heated cutter along a sealing line **44**. Multiple sheets **38** may be stacked on top of each other, preferably with the ends of adjacent sheets **38** separated by spacers. The end of the set of sheets **38** is dipped in a potting mold **46** filled with a potting resin **48**. The potting resin **48** may be, for example, a polyurethane resin formulated to penetrate into the yarns **8** to seal around the various filaments of the cord **10**.

(40) Referring to FIG. **7**, the set of sheets **38** is removed from the potting mold **46** after the potting resin **48** is cured. To expose open ends of the gas transfer membranes **14**, the potting resin **48** is cut through along cutting line **50**. The other end of the set of sheets **38** may be potted in the same manner. Optionally, one or both of the blocks of potting resin **48** may be cut to expose open ends of the gas transfer membranes **14**.

(41) Referring to FIG. **8**, a header **60** is formed by sealing the block of potting resin **48** to a header pan **52**. The header pan **52** may be made of molded plastic and has an outlet **54**. The block of potting resin **48** may be held in the header pan **52** by an adhesive or a gasket **56** between the perimeter of the potting resin **48** and the header pan **52**. Optionally, a second potting material **58** may be pored over the potting resin **48**. The second potting material may further seal the cords **10** or the potting resin **48** to the header pan **52**, or may cushion the cords **10** where they exit from the header **60**. A similar header **60** may be made at the other end of the set of sheets **38**.

(42) Referring to FIG. **9**, a module **66** has two headers **60** with cords **10** extending between them. The headers **60** are preferably vertically aligned and held apart by a frame **62**. The length of the cords **10** between opposing faces of the headers **60** may be slightly greater than the distance between the opposed faces of the headers **60**. In this case the cords **10** have some slack and can sway. The cords **10** are preferably not connected to each other between the headers **60**. Although one cord **10** may contact another as it sways, the movement of a cord **10** is generally independent of other cords **10**. Multiple modules **60** may be held in a common frame **62**. The frame **62** may also hold an aerator **68** near the bottom of a module **66**.

(43) When used for wastewater treatment, the cords **10** are immersed in a bioreactor and a gas, for example air, oxygen, hydrogen or another gas, is fed through the lumen of the gas transfer membranes **14**. A biofilm develops on the outside surface of the cords **10**, and anchors itself by filling the gaps between filaments. The resulting membrane biofilm assembly has a generally circular cross section. The cross section of the membrane biofilm assembly has a diameter of about 0.5 to 3 mm determined assuming that the biofilm forms a film extending no more than 0.5 mm beyond the outer diameter of the core **12**. A more typical biofilm thickness is in the range of 0.05 to 0.2 mm. The sum of the circumferences of the gas transfer membranes **14** multiplied by the length of the cord approximates the active gas transfer surface area while the circumference of the outside of the biofilm multiplied by the length of the cord **10** gives the biofilm area. The sum of the circumferences of the gas transfer membranes **14** is preferably at least 1.5 times the circumference of a circle having the outside diameter of the cord **10**. The sum of the circumferences of the gas transfer membranes **14** is also preferably at least 1.5 times the circumference of the attached biofilm when in use.

(44) Modules of the cords **10** may be deployed in a membrane biofilm reactor (MBfR) by immersing them in an open tank in manner similar to the use of ZeeWeed 500 immersed hollow fibre filtering membranes.

Although the cords **10** will be used to support and transport gas to a biofilm and not for filtration, various system design and operating features of the ZeeWeed 500 system can be adapted. As mentioned above, the cords **10** may be potted with an orderly spacing between them. The module configuration with two headers may be used but modified to use one header of each element for introducing the fresh gas and one header for venting exhausted gas. ZeeWeed cassette frames may be used to facilitate deploying multiple modules with the cords **10** oriented vertically into open tanks. Gas sparging by way of bubbles produced below or near the bottom of the modules can be provided at a low rate to renew the liquid around the cords **10**. Gas sparging at a higher rate may be used to help control biofilm thickness either by the direct action of bubbles, bubble wakes or bubble pressure effects on the biofilm, or by causing cords **10** mounted with slack between the headers to sway in the water to produce turbulence or contact between cords **10**. Optionally, gas exhausted from the cords may be recycled for use in gas sparging.

(45) Referring to FIG. **10**, a module **66** is immersed in a tank **70**. The tank **70** is filled with water to be treated from an inlet **72**. Treated water is removed through an outlet **74**. Optionally, water may recirculate from the outlet **74** to the inlet **72** to provide a flow of water through the module **66**, mix the tank **70**, or to maintain desired conditions in the tank **70**. Air, or another gas, is blown into, or drawn out of, the module **66** by a process gas blower **76**. In the example shown, the gas is blown into one header **60**, travels through the cords **10**, and exhausted from the other header **60**. A throttle valve **78** may be used to increase the gas pressure in the cords **10**. A sparging gas blower **80** blows air or recycled exhaust gas from the module **66**, or both, to the aerator **68** when required for mixing the tank **70** or controlling the thickness of the biofilm on the cords **10**.

(46) Optionally, the aerator **68** may comprise a supply pipe **82** and a transducer **84**. The transducer **84** collects gas ejected from the supply pipe in a pocket below a shell **86**. The pocket of gas grows larger as gas is accumulated as shown in the first two compartments of the shell **86**, counting from the left side of the shell **86**. When the pocket of gas extends to the bottom of J shaped tube, as in the third compartment of the shell **86**, the gas is released through the J shaped tube as shown in the last compartment of the shell **86**. In this way, large bursts of bubbles are released periodically without requiring a large volume of gas to be continuously pumped into the tank **70**. Excessive scouring gas consumes energy and may disturb desirable anoxic or anaerobic conditions in the tank **70**. Periodic large bursts of bubbles can be more effective for renewing the water around the cords **10** or removing biofilm from the cords **10** than the same amount of gas supplied as a continuous stream of bubbles.

(47) In some prior MBfRs, silicon rubber or polydimethylsiloxane (PDMS) are coated over a flat substrate to make a flat sheet membrane. While silicon and PDMS are highly permeable to oxygen, such a flat sheet membrane can provide a surface area for oxygen transfer to surface area of biofilm ratio of only about 1. Further, with reasonably large sheets it is difficult to renew water to be treated along the edges of the sheet or remove excess biofilm from the edges of the sheet. Accordingly, the sheets are often separation by a substantial spacing and the total biofilm area in a tank may be low.

(48) In U.S. Pat. No. 7,169,295 a membrane supported biofilm reactor has modules made with fine hollow fiber membranes. The fine hollow fibers have a thin wall which allows for good gas transfer efficiency even when heat spun polymers are used. However, the fine hollow fibers are also easily damaged. A tow module described in U.S. Pat. No. 7,169,295, although useful in some applications, has loose and exposed hollow fibers which are prone to damage and to being clumped together by biofilm in other applications. Sheet modules described in U.S. Pat. No. 7,169,295 are more resilient and can provide a surface area for oxygen transfer to surface area of biofilm ratio of more than 1. However, like silicon flat sheet module, these sheet modules are still subject to total biofilm area limitations.

(49) The cords **10** described above provide a useful alternative gas transfer module configuration. The use of fine hollow fiber gas transfer membranes **14** allows for good gas transfer efficiencies even when using melt spun polymers and a surface area for oxygen transfer to surface area of biofilm ratio of more than 1. The fine hollow fiber membranes are not loose and exposed. Yet since the cords **10** can move generally independently and do not form a solid sheet, fresh liquid and bubbles used to scour the biofilm to control its thickness can reach cords **10** located in the interior of a module. Movement of the cords **10**, or contact between cords **10**, may also help control biofilm thickness. Further, the total biofilm surface area can be increased relative to a sheet form module.

(50) In a calculated example, a cord **10** comprises a core **12** and two wraps **18**, one in each direction, of gas transfer membrane yarns **15**. The outside diameter of the cord is 1 mm. The wrapping pitch is 5 core diameters resulting in a wrap **18** helix length of 1.18 times the cord length. The cord **10** has 57 meters of 70 micron outside diameter PMP gas transfer membrane **14** per meter of cord **10** length. The surface area of the gas

transfer membranes per unit length of cord is 3.04 times the outer surface area of the cord, calculated based on the circumference of a 1 mm circle.

(51) A biofilm on the cord **10** is assumed to have a thickness of 0.2 mm giving a membrane biofilm assembly diameter of 1.4. The cords **10** are potted in a module in a rectilinear grid with a 0.7 mm gap between their outside surfaces. With biofilm attached, the gap between adjacent cords in a line is 0.4 mm. Using ZeeWeed module moldings, a module **66** has 16 rows of 340 cords **10** each, or 5440 cords **10**. The exposed length of each cord is 1.9 m giving a biofilm area per module of 45.5 square meters. Using a ZeeWeed frame, a cassette with a 3.7 square meter footprint and 2.5 meter height has 64 modules **66** and a total biofilm surface area of 2910 square meters. The biofilm surface area is 315 square meters per cubic meter of cassette volume and 786 square meters per square meter of cassette footprint.

(52) In comparison, a comparable sheet form module made with similar gas transfer membranes **14** can have a 1 mm thick fabric with a similar gas transfer surface area to biofilm surface area of 3.34. The sheets are made into a module with horizontally apposed vertical headers. The module is 2 m long with 1.8 m of exposed membrane length. The sheets and module are 1 m high. The module is 0.3 m wide and can be operated in a 1.5 m deep tank. The centre-to-centre spacing between adjacent sheets is 8 mm. The biofilm surface area is 250 square meters per cubic meter of cassette volume and 250 square meters per square meter of cassette footprint.

(53) As illustrated by the comparison above, the cord **10** module can have more biofilm area per unit volume of module than a sheet form module. Further, tall sheet modules that rely on bubbles for liquid renewal or scouring have been known in the context of filtering membranes to be prone to a chimney effect whereby bubbles and liquid flow are concentrate near the vertical midline of the sheets. This limits the height of sheet modules. It is expected that a cord **10** module can be higher without a similar chimney effect which allows for an additional decrease in tank footprint and land consumption per unit biofilm area.

(54) This written description uses examples to disclose the invention and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art.

Claims

1. A process comprising the steps of, immersing a gas transfer membrane in water; flowing a gas into the gas transfer membrane; supporting a biofilm on the gas transfer membrane; transferring oxygen to the biofilm; withdrawing an exhaust gas from the gas transfer membrane; and, producing bubbles comprising the exhaust gas in the water, wherein the exhaust gas is transferred from the gas transfer membrane to an aerator and wherein the bubbles comprising the exhaust gas are produced by the aerator and wherein the bubbles help control a thickness of the biofilm.
2. The process of claim 1 wherein the exhaust gas is pumped from the gas transfer membrane to the aerator.
3. The process of claim 1 wherein exhaust gas from a plurality of gas transfer membranes is collected in a header before being transferred from-to the aerator.
4. The process of claim 1 wherein the bubbles are produced periodically.
5. The process of claim 1 wherein the aerator comprises a transducer.